UNIT-2 (Part-2)

1. Basic Concepts

Knowledge Representation (KR):

- The area of artificial intelligence (AI) dedicated to representing information about the world in a form that a computer system can utilize to solve complex tasks such as diagnosing a medical condition, understanding natural language, and playing chess.

Key Elements of KR:

- **Symbols:** The basic units of KR, which can represent objects, actions, and events.
- **Syntax:** The rules that govern the structure of expressions in the representation language.
- **Semantics:** The meaning of the symbols and expressions.
- **Inference:** The process of deriving new knowledge from existing knowledge.

Goals of KR:

- **Expressiveness:** The ability to represent a wide range of knowledge.
- **Computational Efficiency:** The ability to compute answers to questions quickly.
- **Understandability:** The ease with which humans can read and understand the representations.

2. Knowledge Representation Paradigms

Paradigms:

1. **Logical Representation:**
    - Uses formal logic to represent knowledge.
    - Types include propositional logic and predicate logic.
2. **Procedural Representation:**
    - Knowledge is represented through procedures or sequences of operations.
    - Example: if-then rules.
3. **Network Representation:**
    - Uses graphs to represent knowledge.
    - Examples include semantic networks and conceptual graphs.
4. **Structured Representation:**
    - Uses structured formats like frames and scripts to represent stereotyped situations.
    - Allows for the organization of knowledge into hierarchies.

3. Propositional Logic

Definition:

- A branch of logic where statements are declared true or false.

Components:

- **Propositions:** Statements that can be true or false.
- **Logical Connectives:** Operators used to form complex sentences (AND, OR, NOT, IMPLIES).

Syntax and Semantics:

- Syntax refers to the symbols and rules used to construct sentences.
- Semantics refers to the meaning assigned to these sentences.

Example:

- p : "It is raining."
- q : "I will carry an umbrella."
- $p \rightarrow q$: "If it is raining, then I will carry an umbrella."

```
p: "It is raining"
q: "I will carry an umbrella"

p → q
```

4. Inference Rules in Propositional Logic

Common Inference Rules:

1. **Modus Ponens:**
   - If $p \rightarrow q$ and p are true, then q is true.
   - Example: If "If it rains, the ground will be wet" is true and "It rains" is true, then "The ground will be wet" is true.
2. **Modus Tollens:**
   - If $p \rightarrow q$ and $\neg q$ are true, then $\neg p$ is true.
   - Example: If "If it rains, the ground will be wet" is true and "The ground is not wet" is true, then "It does not rain" is true.
3. **Disjunction Introduction:**
   - If p is true, then $p \vee q$ is true.
4. **Disjunction Elimination:**
   - If $p \vee q$ is true and $\neg p$ is true, then q is true.
5. **Conjunction:**
   - If p and q are true, then $p \wedge q$ is true.

```
Modus Ponens:
p → q
p
-----
q

Modus Tollens:
p → q
¬q
-----
¬p

Disjunction Introduction:
p
-----
p ∨ q

Disjunction Elimination:
p ∨ q
¬p
-----
q

Conjunction:
p
q
-----
p ∧ q
```

5. Knowledge Representation using Predicate Logic

Predicate Logic:

- Extends propositional logic by dealing with predicates, which can be true or false depending on the arguments.
- Allows for the representation of more complex statements about objects and their relationships.

Components:

- **Predicates:** Functions that return true or false (e.g., Likes(x,y)).
- **Quantifiers:** Symbols that express the quantity of specimens in the domain of discourse (Universal ∀, Existential ∃).
- **Variables:** Symbols representing objects in the domain.

Example:

- $\forall x(Person(x) \rightarrow Mortal(x))$
- "All persons are mortal."

```
∀x ( Person(x) → Mortal(x) )
```

6. Predicate Calculus

Definition:

- A formal system in which one can express statements about properties of objects and the relationships between them using predicates and quantifiers.

Components:

- **Terms:** Variables, constants, and functions.
- **Atomic Formulas:** Predicate symbols applied to terms.
- **Formulas:** Constructed using logical connectives and quantifiers.

Example:

- $\forall x(\text{Student}(x) \rightarrow \exists y(\text{Course}(y) \land \text{Enrolled}(x,y)))$
- "Every student is enrolled in some course."

```
∀x ( Student(x) → ∃y ( Course(y) ∧ Enrolled(x, y) ) )
```

7. Predicate and Arguments

Predicates:

- Describe properties or relations between objects.
- Example: Brother(x,y)

Arguments:

- The objects to which predicates apply.
- Example: In Brother(x,y) x and y are arguments.

Example Statement:

- Likes(Alice,IceCream)
- Predicate: Likes
- Arguments: Alice, IceCream

```
Likes(Alice, IceCream)
```

8. ISA Hierarchy

Definition:

- A hierarchical representation of knowledge where higher-level concepts (more general) include lower-level concepts (more specific).

Components:

- **Classes:** General categories (e.g., Animal).
- **Subclasses:** Specific instances of classes (e.g., Dog, Cat).

Example:

- Animal → Mammal → Dog → Labrador

```
Animal
  |
  +-- Mammal
        |
        +-- Dog
              |
              +-- Labrador
```

9. Frame Notation

Definition:

- A data structure for representing a stereotyped situation, like a frame in a play.

Components:

- **Slots:** Attributes or properties of the frame.
- **Values:** The specific data for each slot.

Example:

- Frame for "Restaurant Visit":
  - Slots: Restaurant-Name, Time, Food-Ordered
  - Values: "Joe's Diner", "7 PM", "Pizza"

```
Frame: RestaurantVisit
  Restaurant-Name: "Joe's Diner"
  Time: "7 PM"
  Food-Ordered: "Pizza"
```

10. Resolution

Definition:

- A rule of inference used for automated theorem proving in predicate logic and propositional logic.

Process:

- Convert all statements to a standard form (Conjunctive Normal Form).
- Apply the resolution rule to derive a contradiction.

Example:

- ¬A∨B, A∨C
- Resolution: B∨C

```
¬A ∨ B
A ∨ C
---------
B ∨ C
```

11. Natural Deduction

Definition:

- A method for deriving logical conclusions from premises using a set of inference rules.

Process:

- Start with premises and apply inference rules to derive conclusions.

Common Rules:

- **Introduction and Elimination Rules** for each logical connective (AND, OR, NOT, IMPLIES).

Example:

- Given A→B and A, deduce B(using Modus Ponens).

```
A → B
A
-----
B
```